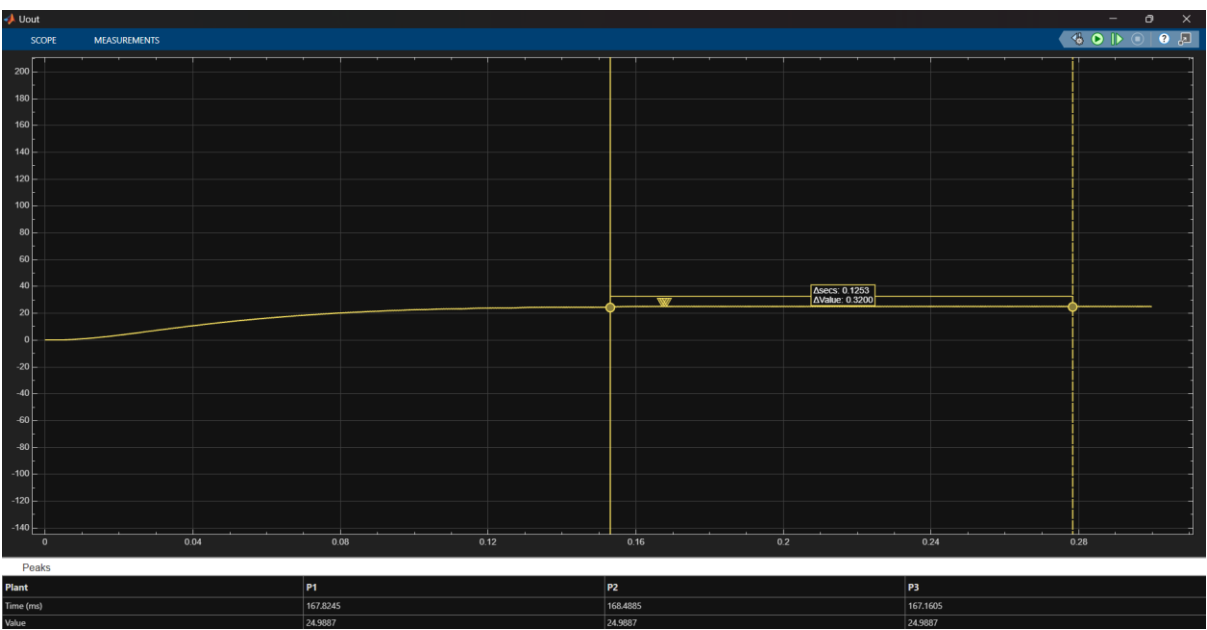
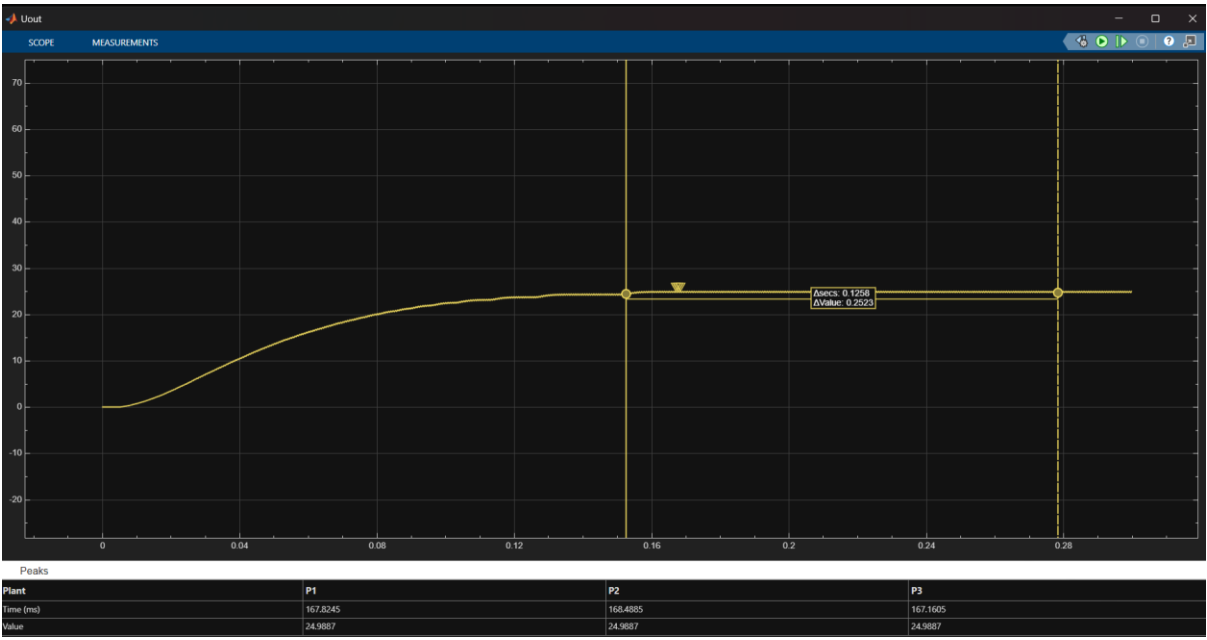


PE PROJECT PART 5

Filtering factor – 0.7



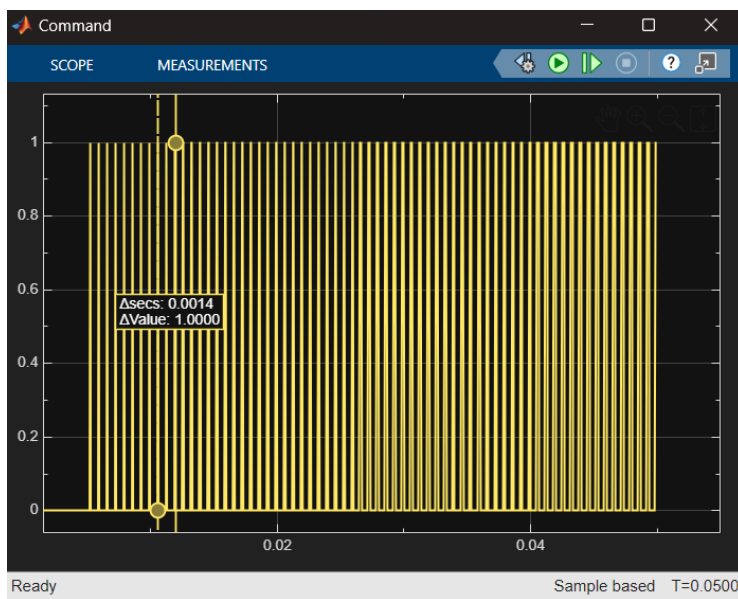
(it also shows me like this one below if i restart simulink because it keeps crashing, but the settling time should be pretty much the same or very similar)





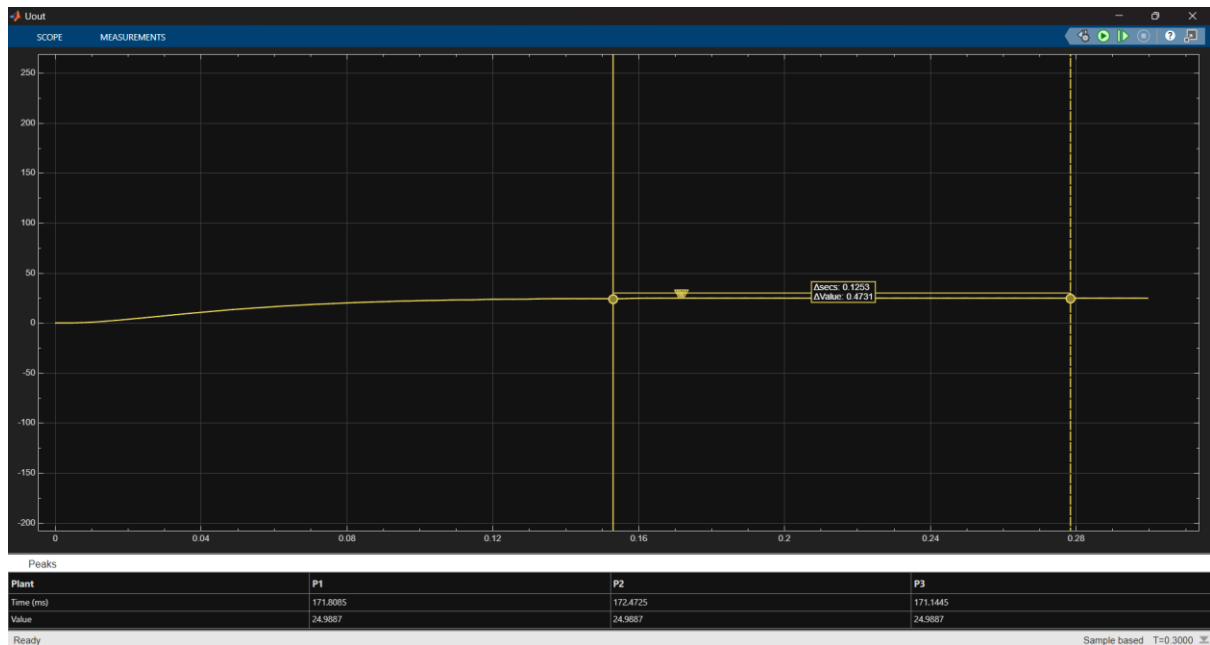
- Settling time – 0.1525 s
- Final value - 24.9887 v
- Smooth response, balanced filtering
- Less filtering - more responsive to input changes
- Stable, Reaches close to 25V (the setpoint)

This ripple is caused by PWM switching.



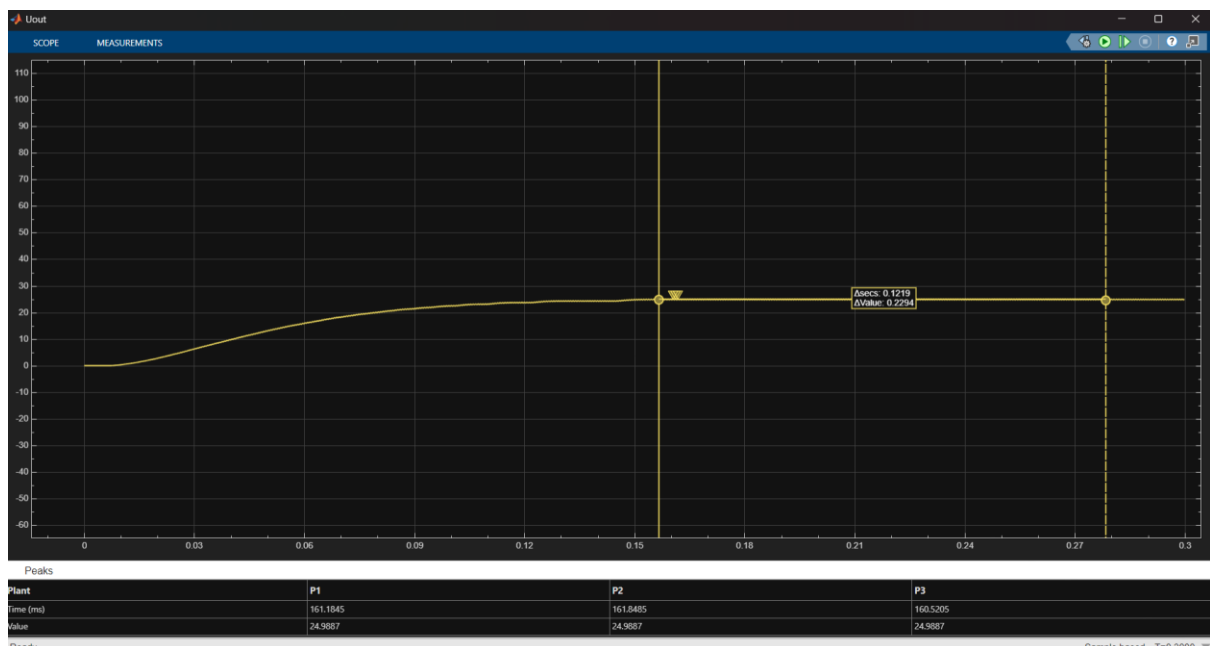
The command signal = PWM duty cycle, switching between 0 and 1.

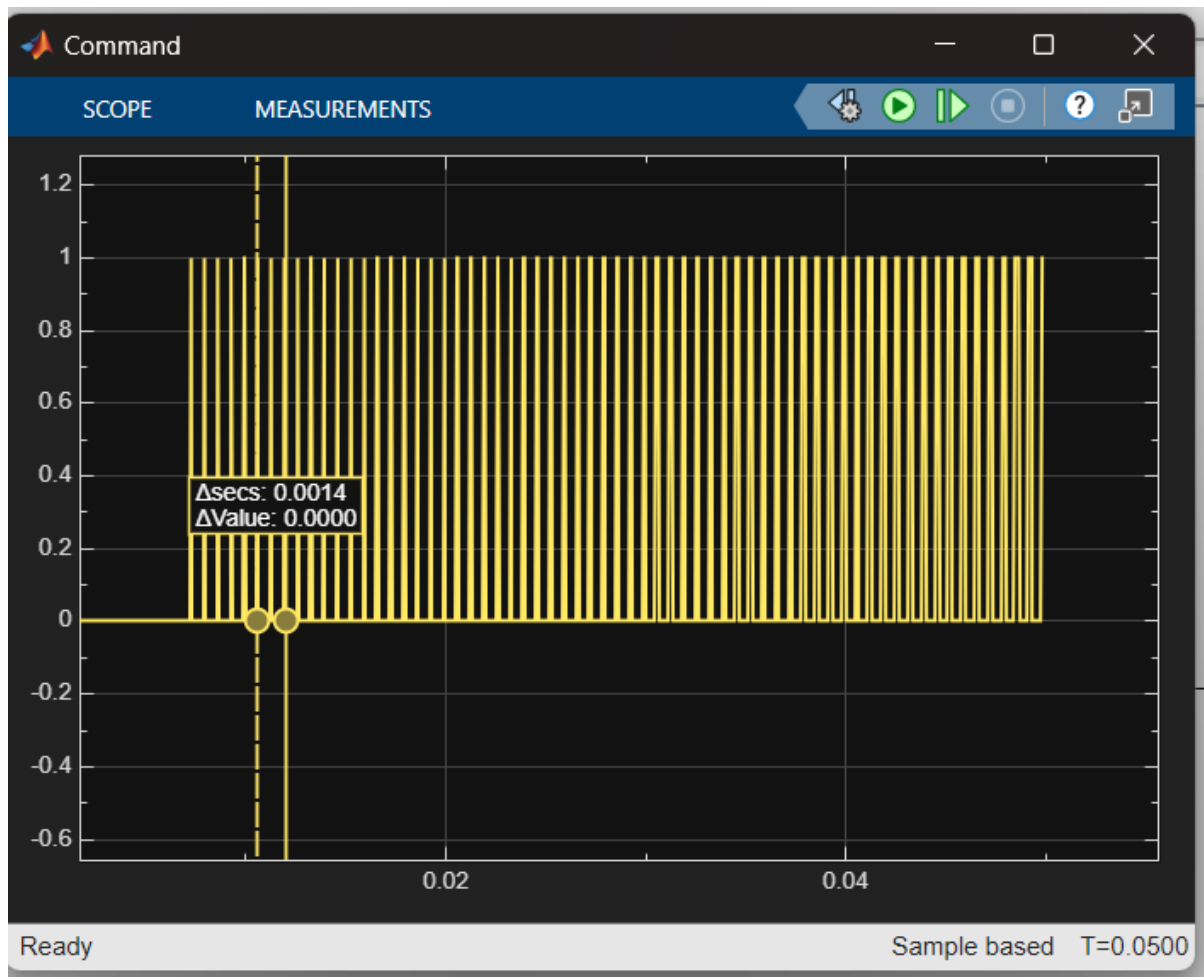
Filtering factor – 0.1 a lot of filtering



- Settling time:  $\sim 0.1565$  s
- Peak – 24.9887 V
- More filtering – slower response, causes a bit of lag

Filtering factor – 0.9 (minimal filtering)





- Settling time:  $\sim 0.1459$
- Peak is the same
- Quickest response